Small rational lifts of reciprocal character sums at 256-bit primes: an empirical torsion dichotomy

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Abstract

For a squarefree cubic or quartic f over $\mathbb{Q}$ and a rational pole a, the reciprocal character sum $C_f(a) = \sum_{x \neq a} \chi(f(x))/(x-a) \in \mathbb{F}_p$ can be evaluated with one differentiated Miller chain once the trace of Frobenius is known. When the associated twist point R_a has finite order over $\mathbb{Q}$, a torsion-collapse formula gives $C_f(a) \equiv A\varepsilon + B\tau \pmod p$ for fixed rationals A, B, where $\varepsilon = \chi(f(a))$ and τ is the trace, so the residue is the reduction of a rational number u/d with $|u|d = O(\sqrt{p})$. We ask whether anything else produces such small rational lifts. They can be detected without a basis of guessed quantities, from the continued fraction of $C_f(a)/p$, and at primes of cryptographic size, where the Miller-chain evaluator still runs in milliseconds, a single evaluation gives strong discrimination against a uniform-residue model. We report a pre-registered survey of 2,668 pairs (f, a) over 184 curves, most without complex multiplication, each evaluated at three random 256-bit primes. Small lifts occurred at exactly the 80 pairs with R_a torsion (orders 2, 3, 5 and 6), whose recovered relations equalled the predicted (A, B), and at none of the 2,588 others; for nonzero residues the minimal lift sizes of the two groups were separated by more than 100 bits. Every torsion relation is *balanced*, meaning $A + B = g_1/2$ with $g_1 = f'(a)/f(a)$. We prove a uniform converse for balanced relations: if one holds at all but finitely many admissible primes, then R_a is torsion. The proof combines a Cartier-operator form of the curve-order identity with Bost's theorem on the Grothendieck–Katz conjecture for solvable groups. The general unbalanced uniform converse is not established here. The density-zero converse that we state as a conjecture would follow from an equidistribution conjecture of Katz; the survey is consistent with it but cannot test an asymptotic statement.

1 Introduction

Let $p \geq 5$ be prime, χ the quadratic character of $\mathbb{F}_p$, and $f \in \mathbb{F}_p[x]$ a squarefree cubic or quartic. For $a \in \mathbb{F}_p$ with $f(a) \neq 0$ put

$$C_f(a) = \sum_{x \in \mathbb{F}_p,\ x \neq a} \frac{\chi(f(x))}{x - a} \in \mathbb{F}_p. \tag{1.1}$$

Theorem B of Jordan (2026) evaluates $C_f(a)$ with one differentiated Miller chain (Miller, 2004) of length $O(\log p)$ on a quadratic twist, once the trace of Frobenius τ_f of $v^2 = f(x)$ is known; the trace is computed once per curve and serves every pole a. When the twist point R_a is torsion, the chain collapses and $C_f(a)$ becomes a fixed rational combination of $\varepsilon = \chi(f(a))$ and τ_f (Jordan, 2026, Proposition 6.2). Theorem 6.3 of that paper, a closed form for $\sum_{k \leq p/3} \binom{2k}{k} \bmod p$, arises this way.

This note asks whether torsion is the only source of such closed forms. Any closed form of that shape makes $C_f(a)$ the reduction of a rational number u/d with $|u|d = O(\sqrt{p})$, whereas few residues have so small a representation (Lemma 3.3). The question can therefore be examined

without guessing which p-dependent quantities might appear, by measuring the smallest $|u|d$ with $C_f(a) \equiv u/d$. Section 3 makes this precise with continued fractions.

Part of the converse can be proved. Every relation produced by torsion is balanced, in the sense that $A + B = g_1/2$, which is equivalent to $\alpha_N(R_a) \equiv c\,N \pmod p$ for a fixed constant c. Theorem 2.5 shows that if a balanced relation holds at all but finitely many admissible primes, then R_a is torsion. The argument views $\alpha_N(R_a)$ through the Cartier operator, as the obstruction to a rank-one connection on E_a having vanishing p-curvature, and then applies Bost's theorem (Bost, 2001). It says nothing about relations holding only on a set of primes of positive density, about unbalanced relations, or about lifts outside the span of ε and τ_f; the survey addresses the last two empirically, at individual large primes.

The discrimination this gives grows with p. By Lemma 3.3, a uniform residue has a lift with $|u|d \le 10\sqrt{p}$ with probability at most about 0.2 at $p \approx 10^6$, so a lift of that size would have to recur over many primes before it stood out, but with probability below 10^{-35} at $p \approx 2^{256}$, where one evaluation gives strong discrimination against a uniform-residue model. At such primes direct summation of (1.1) needs about p terms, and the standard Bostan–Gaudry–Schost recurrence route (Bostan, Gaudry & Schost, 2007) about 2^{128} field operations per value, with logarithmic factors suppressed; the Miller chain needs a few hundred curve operations. The survey reported here comprised 8,004 evaluations at 256-bit primes, which took 83 s in pure Python on a laptop after 552 point counts that took 1,151 s.

Section 2 recalls the evaluator and the known direction and proves the balanced converse, Section 3 gives the size test, Section 4 the pre-registered design, Section 5 the results and Section 6 the limitations.

2 Setting and the known direction

Let $f \in \mathbb{Q}[x]$ be a squarefree cubic or quartic and $a \in \mathbb{Q}$ with $f(a) \neq 0$. Write

$$g(z) = \frac{f(a+z)}{f(a)} = 1 + g_1 z + g_2 z^2 + g_3 z^3 + g_4 z^4, \tag{2.1}$$

with $g_4 = 0$ for a cubic. Taking $(0,1)$ as origin on $v^2 = g(z)$ gives the short model $E_a : Y^2 = X^3 + A_g X + B_g$, and $(0,-1)$ maps to the point

$$R_a = \Big(\tfrac{1}{4}g_1^2 - \tfrac{2}{3}g_2,\ -g_3 + \tfrac{1}{2}g_1 g_2 - \tfrac{1}{8}g_1^3\Big) \in E_a(\mathbb{Q}), \tag{2.2}$$

with A_g, B_g as in Jordan (2026, equation (4.2)). Call a prime p *admissible* for (f,a) if $p \ge 5$, the coefficients of f and a are p-integral, the leading coefficient and discriminant of f are p-adic units, and $f(a) \not\equiv 0 \pmod p$. For admissible p put $\varepsilon = \chi(f(a))$ and $N = p + 1 - \varepsilon\tau_f = \#E_a(\mathbb{F}_p)$. On E_a use the uniformizer $t = -X/Y$ at the origin O. For a point P and $m \ge 1$, let $f_{m,P}$ be a Miller function with $\operatorname{div}(f_{m,P}) = m(P) - ([m]P) - (m-1)(O)$, and put

$$\alpha_m(P) = [t^0]\,\frac{d \log f_{m,P}}{dt}, \tag{2.3}$$

the constant Laurent coefficient of its logarithmic derivative at O, as in Jordan (2026, equation (1.3)). It is independent of the scalar normalization of $f_{m,P}$ and is computed by the differentiated Miller chain.

Theorem 2.1 (Jordan, 2026, Theorem B)**.** *For every admissible prime p, $C_f(a) = \varepsilon\big(-\frac{g_1}{4}(N-2) - \frac{1}{2}\alpha_N(R_a)\big)$ in $\mathbb{F}_p$.*

Proposition 2.2 (torsion implies a closed form)**.** *Suppose R_a has finite order m in $E_a(\mathbb{Q})$, and let $\alpha_m(R_a) \in \mathbb{Q}$ be computed by the Miller chain over $\mathbb{Q}$. Put*

$$A = \frac{g_1}{4} - \frac{\alpha_m(R_a)}{2m}, \qquad B = \frac{g_1}{4} + \frac{\alpha_m(R_a)}{2m}. \tag{2.4}$$

Then for every admissible prime p with $p \nmid m$ and A, B, g_1 p-integral,

$$C_f(a) \equiv A\,\varepsilon + B\,\tau_f \pmod{p}. \tag{2.5}$$

Proof. At an admissible prime E_a has good reduction, and reduction is injective on torsion of order prime to p, so R_a has order exactly m in $E_a(\mathbb{F}_p)$ and the Miller chain for α_m reduces modulo p step by step. By Jordan (2026, Proposition 6.2), $\alpha_N(R_a) = q\,\alpha_m(R_a)$ with $q = N/m$, and $N \equiv 1 - \varepsilon\tau_f \pmod{p}$. Substituting into Theorem 2.1 and using $\varepsilon^2 = 1$,

$$C_f(a) \equiv \varepsilon\Big(\frac{g_1}{4}(1+\varepsilon\tau_f) - \frac{(1-\varepsilon\tau_f)\,\alpha_m}{2m}\Big) = \Big(\frac{g_1}{4} - \frac{\alpha_m}{2m}\Big)\varepsilon + \Big(\frac{g_1}{4} + \frac{\alpha_m}{2m}\Big)\tau_f, \tag{2.6}$$

which is (2.5) with A, B as in (2.4). □

Since $|\tau_f| \le 2\sqrt{p}$ by Hasse's bound, clearing the denominator D of A and B shows that $C_f(a) \equiv u/D$ with $|u| = O(\sqrt{p})$.

Definition 2.3. A relation (2.5) with fixed $A, B \in \mathbb{Q}$ is *balanced* if $A + B = g_1/2$.

Every relation supplied by Proposition 2.2 is balanced. At an admissible prime, Theorem 2.1, $\varepsilon^2 = 1$ and $N - 2 \equiv -1 - \varepsilon\tau_f$ show that (2.5) is equivalent to

$$\alpha_N(R_a) \equiv \Big(\frac{g_1}{2} - 2A\Big) + \Big(\frac{g_1}{2} - 2B\Big)\varepsilon\tau_f \pmod{p}. \tag{2.7}$$

Since $N \equiv 1 - \varepsilon\tau_f$, a balanced relation is therefore exactly a congruence $\alpha_N(R_a) \equiv c\,N \pmod{p}$ with the fixed constant $c = B - A$; in the torsion case $c = \alpha_m(R_a)/m$.

2.1 A uniform converse for balanced relations

Write $\mathcal{C}_p$ for the Cartier operator on differentials in characteristic p. It is p^{-1}-linear, fixes logarithmic differentials dg/g, and satisfies $\mathcal{C}_p\big(\sum_n b_n X^n\,dX\big) = \sum_j b_{pj+p-1}^{1/p} X^j\,dX$ (Cartier, 1957; Jordan, 2026, Section 2.2).

Lemma 2.4 (Cartier form of the curve-order identity)**.** *Let $p \ge 5$, let $W \in \mathbb{F}_p[X]$ be a monic squarefree cubic and $E : Y^2 = W(X)$ the corresponding elliptic curve, with $N = \#E(\mathbb{F}_p)$, and let $P = (r, s) \in E(\mathbb{F}_p)$ with $s \ne 0$. Put $\omega_0 = dX/(2Y)$ and*

$$\vartheta_P = \frac{Y + s}{X - r}\,\omega_0, \tag{2.8}$$

a differential with simple poles at P and O only, of residues $+1$ and -1. Then

$$\mathcal{C}_p(\omega_0) = H_E\,\omega_0, \qquad \mathcal{C}_p(\vartheta_P) - \vartheta_P = \alpha_N(P)\,\omega_0, \tag{2.9}$$

where $H_E = [X^{p-1}]W^{(p-1)/2}$ is the Hasse invariant, so that $N \equiv 1 - H_E \pmod{p}$.

Proof. Put $h = (p-1)/2$ and write $W^h = U + H_E X^{p-1} + X^p J_W$ with $\deg U \le p - 2$, as in Jordan (2026, Theorem A). Since $dX/Y = W^h\,dX/Y^p$ and $\mathcal{C}_p$ is p^{-1}-linear, $\mathcal{C}_p(g\,dX/Y) = \mathcal{C}_p(gW^h\,dX)/Y$ for $g \in \mathbb{F}_p(X)$; with $g = 1$ this gives $\mathcal{C}_p(\omega_0) = H_E\omega_0$, and $H_E \equiv \tau_E = p + 1 - N$. Next,

$$\vartheta_P = \frac{1}{2}\,\frac{dX}{X - r} + \frac{s}{2}\,\frac{dX}{(X-r)Y}. \tag{2.10}$$

The first term of (2.10) is half a logarithmic differential, and $\frac{1}{2} \in \mathbb{F}_p$, so $\mathcal{C}_p$ fixes it. For the second, $W^h/(X-r) = W^h(r)/(X-r) + q(X)$ with $q = (W^h - W^h(r))/(X-r)$ a polynomial of degree $3h - 1 < 2p - 1$, and $W^h(r) = s^{p-1} = 1$. The operator fixes $dX/(X-r)$, and

$\mathcal{C}_p(q\,dX) = [X^{p-1}]q\;dX = J_W(r)\,dX$, because $[X^{p-1}]\,(X^k - r^k)/(X-r) = r^{k-p}$ for $k \geq p$ and vanishes for $k < p$. Hence

$$\mathcal{C}_p\Big(\frac{s}{2}\,\frac{dX}{(X-r)Y}\Big) = \frac{s}{2}\Big(\frac{1}{X-r} + J_W(r)\Big)\frac{dX}{Y}, \tag{2.11}$$

and $\mathcal{C}_p(\vartheta_P) - \vartheta_P = s\,J_W(r)\,\omega_0 = \alpha_N(P)\,\omega_0$ by Theorem A of Jordan (2026), which holds for every N, including $p \mid N$. □

Theorem 2.5 (uniform converse for balanced relations)**.** *Let $f \in \mathbb{Q}[x]$ be a squarefree cubic or quartic, $a \in \mathbb{Q}$ with $f(a) \neq 0$, and $A, B \in \mathbb{Q}$ with $A + B = g_1/2$. If (2.5) holds at all but finitely many admissible primes, then R_a has finite order in $E_a(\mathbb{Q})$. Consequently, by Proposition 2.2, R_a is torsion if and only if some balanced relation holds at all but finitely many admissible primes.*

Proof. Write $E = E_a$ and $P = R_a = (r, s)$. If $s = 0$, then P has order 2. Otherwise put $c = B - A$ and $\eta = \vartheta_P + c\,\omega_0$, with ϑ_P as in (2.8), a rational differential on E defined over $\mathbb{Q}$, with simple poles at P and O only, of residues $+1$ and -1. Let $L = \mathcal{O}_E\big(-(P) + (O)\big)$ and $\nabla\varphi = d\varphi - \varphi\,\eta$. At P, a local generator of L is a uniformizer π, and $\nabla\pi = -\pi\,(\eta - d\pi/\pi)$ with $\eta - d\pi/\pi$ regular; at O the generator π^{-1} behaves in the same way, since the residue there is -1. So ∇ is a regular connection on L, flat because E is a curve. The pair (L, ∇) is defined over $\mathbb{Q}$ and extends to a smooth model of E over $\mathbb{Z}[1/M]$ for a suitable integer M.

Let $p \nmid M$ be an admissible prime at which (2.5) holds. By (2.7), $\alpha_N(P) \equiv cN$, and (2.9) gives, in characteristic p,

$$(\mathcal{C}_p - 1)\,\eta = \big(\alpha_N(P) + c\,(H_E - 1)\big)\,\omega_0 = \big(\alpha_N(P) - cN\big)\,\omega_0 = 0. \tag{2.12}$$

Since $\mathcal{C}_p(\eta) = \eta$ by (2.12), Cartier's theorem on logarithmic differentials gives $\eta = dh/h$ for some h in the function field (Cartier, 1957; Katz, 1970, Section 7). Then $\nabla h = dh - h\eta = 0$, so h is a rational horizontal basis of $d - \eta$, and the p-curvature vanishes at the generic point; compare the Cartier descent for connections of zero p-curvature (Katz, 1970, Theorem 5.1). Since the p-curvature is $\mathcal{O}$-linear and ∇ is regular, it then vanishes everywhere. Thus the reduction of (L, ∇) has vanishing p-curvature at all but finitely many primes.

A line bundle with connection is a principal $\mathbb{G}_m$-bundle with connection, and $\mathbb{G}_m$ is solvable. By Bost (2001, Theorem 2.9), there is a finite étale covering $\pi : Y \to E$ on which (L, ∇) becomes trivial; in particular $\pi^*L \simeq \mathcal{O}_Y$. If π has degree d, then $L^{\otimes d} \simeq \mathrm{Nm}_\pi(\pi^*L) \simeq \mathcal{O}_E$. The class of $(P) - (O)$ is therefore d-torsion in $\mathrm{Pic}^0(E)$, and under $E(\overline{\mathbb{Q}}) \simeq \mathrm{Pic}^0(E_{\overline{\mathbb{Q}}})$ this gives $[d]P = O$. □

Remark 2.6. The theorem is uniform: it needs the relation at all but finitely many primes. For an unbalanced relation, (2.7) gives $\alpha_N(R_a) \equiv cN + c'$ with a fixed $c' \neq 0$, so $(\mathcal{C}_p - 1)\eta = c'\omega_0 \neq 0$ and the argument does not apply; the general unbalanced uniform converse is not established here. The theorem also says nothing about relations holding on a set of primes of positive density, or about small lifts outside the span of ε and τ_f. The argument assembles known results, and its conclusion may be familiar to specialists; it was identified in an AI-assisted literature review and audit, and has not yet been reviewed by a specialist.

Example 2.7. Each of the following was checked by direct summation of (1.1) and a brute-force trace at every admissible prime in the stated range (all primes $7 \leq p < 3000$ for (i)–(iii), and $7 \leq p < 2000$ for (iv); the inadmissible primes $p = 7$ in (ii), where $f(3) = 7$ and the discriminant is 49, and $p = 11$ in (iii), where the discriminant is -44, fall outside the statement). Each is also an instance of Proposition 2.2, with the stated order of R_a computed exactly over $\mathbb{Q}$.

(i) Order 2: $f = x^3 - 2x^2 + 2x - 1$, $a = 0$: $C_f(0) \equiv -\frac{1}{2}(\varepsilon + \tau_f)$.

(ii) Order 3: $f = x^3 - 2x^2 - x + 1$, $a = 3$: $C_f(3) \equiv \frac{1}{3}(2\varepsilon + \tau_f)$.

(iii) Order 5: $f = x^3 - 2x^2 + 2$, $a = 0$: $\displaystyle\sum_{x \neq 0} \frac{\chi(x^3 - 2x^2 + 2)}{x} \equiv \frac{\chi(2) - \tau_f}{5}$.

(iv) Order 6: $f = x^4 - x^2 - 1$, $a = 1$: $C_f(1) \equiv \frac{1}{3}(\tau_f - 4\chi(-1))$.

Ten pairs in the grid of Section 4 have $A = B = 0$, so that $C_f(a) \equiv 0$; for example $f = x^3 + d$ at $a = 0$ (for $p \equiv 1 \pmod 3$ the substitution $x \mapsto \omega x$ by a cube root of unity forces vanishing, and the proposition gives it for all admissible p), and even quartics at $a = 0$, where $x \mapsto -x$ forces it.

Theorem 2.5 settles the uniform converse for balanced relations. A much stronger converse, that for R_a of infinite order no fixed relation (2.5) holds at infinitely many primes, is stronger than any heuristic we can offer: if $C_f(a)$ behaved like a uniform residue, a fixed relation would hold at p with probability about $1/p$, and since $\sum_p 1/p$ diverges, an independence assumption would predict infinitely many accidental coincidences on a set of density zero. We therefore state the converse in density form.

Conjecture 2.8. *Let $f \in \mathbb{Q}[x]$ be a squarefree cubic or quartic and $a \in \mathbb{Q}$ with $f(a) \neq 0$, and suppose R_a has infinite order in $E_a(\mathbb{Q})$. Then for all $A, B \in \mathbb{Q}$, the set of admissible primes p at which (2.5) holds has relative natural density zero among admissible primes.*

This remains conjectural. For balanced relations, Theorem 2.5 shows only that the set of admissible primes at which (2.5) fails is infinite; Conjecture 2.8 asks for the much stronger statement that the set where it holds has density zero, and for unbalanced relations nothing is proved. The survey of Section 4 addresses a different, finite question: whether, at three sampled primes, $C_f(a)$ is the reduction of *any* rational u/d with small $|u|d$. A small lift at a non-torsion pair would point to structure beyond Proposition 2.2. The absence of such lifts is consistent with Conjecture 2.8 but does not bear directly on an asymptotic density statement.

3 A basis-free size test

Definition 3.1. For $x \in \mathbb{F}_p^\times$ represented in $[1, p-1]$, let $x/p = [0; a_1, \ldots, a_n]$ be the canonical finite continued fraction (with $a_n \geq 2$), let $a_{\max} = \max_i a_i$, and define

$$\theta_p(x) = 1 - \frac{\log a_{\max}}{\log p}, \qquad \theta_p(0) = 0. \tag{3.1}$$

If $a_{\max} = a_{k+1}$ and h_k/q_k is the preceding convergent, the associated lift is $x \equiv u/d$ with $d = q_k$ and $u = q_k x - h_k p$. Define also the minimal lift size of $x \in \mathbb{F}_p^\times$ by

$$S_p(x) = \min\bigl\{|u|d : \ d \geq 1, \ p \nmid d, \ u \in \mathbb{Z} \setminus \{0\}, \ u \equiv dx \pmod p\bigr\}, \tag{3.2}$$

and set $S_p(0) = 0$, consistently with $\theta_p(0) = 0$.

Lemma 3.2. *For $x \in \mathbb{F}_p^\times$,*

$$\frac{p}{a_{\max} + 2} < S_p(x) \leq \frac{p}{a_{\max}}, \qquad \textit{hence} \qquad \frac{\log S_p(x)}{\log p} \leq \theta_p(x) < \frac{\log S_p(x) + \log 3}{\log p}. \tag{3.3}$$

Proof. Upper bound. Let $a_{\max} = a_{k+1}$, with $k < n$. Then $1 \leq q_k < q_n = p$, so $p \nmid q_k$, and $x/p \neq h_k/q_k$, so $u = q_k x - h_k p \neq 0$. From $|x/p - h_k/q_k| \leq 1/(q_k q_{k+1})$, with equality when $k = n - 1$, and $q_{k+1} \geq a_{k+1} q_k$ we get $|u| q_k \leq p q_k / q_{k+1} \leq p/a_{k+1}$.

Lower bound. Let (u, d) attain $S_p(x)$. Taking $d = 1$ and u the centred representative of x shows $S_p(x) < p/2$, so $|u|d < p/2$. Write $dx - u = kp$; then $|x/p - k/d| = |u|/(dp) < 1/(2d^2)$. Reduce k/d to lowest terms h/q with $q \leq d$. Then $|x/p - h/q| < 1/(2q^2)$, so h/q is a convergent h_j/q_j by Legendre's criterion (Hardy & Wright, 2008), with $j < n$ because $u \neq 0$. The error

formula $|x/p - h_j/q_j| = 1/\big(q_j(\alpha_{j+1}q_j + q_{j-1})\big)$, where the complete quotient satisfies $\alpha_{j+1} < a_{j+1} + 1$ (or $\alpha_n = a_n$) and $q_{j-1} \le q_j$ (Khinchin, 1964), gives $|x/p - h_j/q_j| > 1/\big(q^2(a_{j+1}+2)\big)$. Hence $|u|d > d^2p/\big(q^2(a_{j+1}+2)\big) \ge p/(a_{\max}+2)$.

The bounds on θ_p follow by taking logarithms and using $a_{\max} + 2 \le 3a_{\max}$. □

Under Proposition 2.2, $S_p(C_f(a)) = O(\sqrt{p})$, so Lemma 3.2 gives $\theta_p(C_f(a)) \le \frac{1}{2} + O(1/\log p)$. The statistic can be smaller still when the trace is small, when the τ coefficient vanishes (then $C_f(a)$ is a fixed rational and $\theta_p \approx 0$), or when an unusually small alternative lift exists; the zero residue has $S_p = \theta_p = 0$ by convention. The registered threshold $\theta_p \le 3/4$ implies $S_p \le p^{3/4}$; conversely $S_p \le p^{3/4}/3$ implies $\theta_p < 3/4$, so the two criteria differ only in a narrow boundary band.

Lemma 3.3. *Let X be uniform on $\mathbb{F}_p^\times$ and $1 \le T < p$. Then*

$$\Pr\big(S_p(X) \le T\big) \le \frac{2}{p-1} \sum_{d=1}^{\lfloor T \rfloor} \Big\lfloor \frac{T}{d} \Big\rfloor \le \frac{2T(1+\log T)}{p-1}. \tag{3.4}$$

Proof. Every x with $S_p(x) \le T$ is u/d for some $1 \le d \le T$ and $1 \le |u| \le T/d$, and each denominator d contributes at most $2\lfloor T/d \rfloor$ numerators; repeated representations only overcount. Finally $\sum_{d \le T} T/d \le T(1 + \log T)$. □

With $T = p^{3/4}$, (3.3) and (3.4) give, for a uniform nonzero residue,

$$\Pr\big(\theta_p \le \tfrac{3}{4}\big) \le \frac{2p^{3/4}\big(1 + \frac{3}{4}\log p\big)}{p-1} \approx 1.5 \cdot 10^{-17} \qquad (\log_2 p \approx 255.6), \tag{3.5}$$

with values 1.48–$1.56 \cdot 10^{-17}$ at the three primes of the survey. A uniform residue on all of $\mathbb{F}_p$ is zero with the additional probability $1/p$. Inequality (3.5) is rigorous for uniform residues; that the sums $C_f(a)$ at non-torsion pairs behave like uniform residues is an unproved modelling assumption.

Remark 3.4. Large primes give much sharper discrimination per evaluation. A torsion lift has $S_p \le C\sqrt{p}$ with C depending on D, A and B, and by Lemma 3.3, when $1 \le C\sqrt{p} < p$, a uniform residue shares this property with probability at most $2C\sqrt{p}\,(1 + \log(C\sqrt{p}))/(p-1)$. At small primes this upper bound is not small and the constant C may be comparable with $\sqrt{p}$, so accidental candidates are common and a lift would have to be confirmed over many primes. At $p \approx 2^{256}$ one evaluation gives strong discrimination against the uniform-residue model, although a single small lift would not by itself establish torsion, a persistent formula or any asymptotic statement. A search restricted to guessed quantities, such as ε and τ_f, remains feasible at small primes by direct summation and a relation search across primes, but it detects only relations among the quantities included. Recovering rationals from residues by size is classical (Wang, Guy & Davenport, 1982).

4 Pre-registered design

All choices below were frozen in a pre-registration file before the run, together with the predicted pair of rationals (A, B) for every torsion pair. The run used fixed seeds and the code was fingerprinted by SHA-256.

Grid. The curves are the squarefree monic cubics $x^3 + bx^2 + cx + d$ with $b, c, d \in \{-2, \dots, 2\}$ (116 curves) and the squarefree monic quartics $x^4 + bx^3 + cx^2 + dx + e$ with $b, c, d, e \in \{-1, 0, 1\}$ (68 curves). The poles are $a \in \{0, \pm 1, \pm 2, \pm 3, \pm\frac{1}{2}, \pm\frac{1}{3}, \pm\frac{3}{2}, \pm\frac{2}{3}\}$ with $f(a) \ne 0$. This gives 2,668 pairs (f, a).

Registered hypothesis and predictions. The order of R_a over $\mathbb{Q}$ was computed exactly, by testing $[m]R_a = O$ for $1 \le m \le 12$ in rational arithmetic; by Mazur's theorem (Mazur, 1977), a point of $E_a(\mathbb{Q})$ that passes no such test has infinite order. Eighty pairs have R_a torsion: 38 of order 2, 26 of order 3, 12 of order 5 and 4 of order 6. For each, (A, B) was computed from Proposition 2.2 and registered. The registered hypothesis (H4) was: *for every registered pair, $C_f(a)$ has a small lift (class LIFT below) if and only if R_a has finite order over $\mathbb{Q}$*, with the predictions that all 80 torsion pairs are LIFT with fitted relation equal to the registered (A, B) and that all other pairs are NONE. This operational hypothesis concerns the sampled primes only; it is distinct from Conjecture 2.8, which concerns asymptotic density and was formulated in this note afterwards.

Primes and traces. Three random 256-bit primes were drawn from a registered seed and used for every pair. For each curve and prime, the trace was obtained from PARI/GP (The PARI Group, n.d.; version 2.19.0) by `ellap` on the twist E_{a_0} at a fixed reference pole a_0, using the sign rule $\tau_g = \varepsilon\tau_f$ of Jordan (2026). As an independent consistency check on each supplied trace, not a certificate of the point count, the run verified $[p + 1 - \tau_g]P = O$ at four random points of $E_{a_0}(\mathbb{F}_p)$, and was to stop on any failure.

Evaluation and classification. For each pair and prime, $C_f(a)$ was evaluated by Theorem 2.1, with the chain required to return $[N]R_a = O$, and θ_p computed by Definition 3.1. A pair was classed LIFT if $\theta_p \le 3/4$ at all three primes, NONE if $\theta_p > 3/4$ at all three, and MIXED otherwise.

Relation recovery. For pairs not classed NONE, with $x_i = C_f(a)$, ε_i and τ_i at the primes p_1, p_2, p_3, an integer vector (D, a_1, a_2) with $D \ne 0$ satisfying

$$Dx_i \equiv a_1\varepsilon_i + a_2\tau_i \pmod{p_i} \qquad (i = 1, 2, 3) \tag{4.1}$$

was sought by lattice reduction modulo $p_1p_2p_3$ (Lenstra, Lenstra & Lovász, 1982), normalized to $D > 0$, and accepted if its entries were at most 10^6 in absolute value. The relation recovered from (4.1) is then $(A, B) = (a_1/D,\ a_2/D)$ in lowest terms, and *recovery of the registered relation* means equality of these rationals with the registered (A, B).

Confirmation and verdict. Non-torsion pairs classed LIFT or MIXED were to be frozen by SHA-256 and retested at five fresh 256-bit primes; a pair LIFT at all five would falsify H4. A torsion pair not classed LIFT, or with a relation other than the registered one, would make the run invalid.

Stage 0. Before the survey the run checked:

(a) that the grid and all 80 predictions, recomputed, equal the registered ones;

(b) the PARI traces, through the same interface and sign rule, against brute-force point counts at the 20 primes $101 \le p \le 197$: 3,678 comparisons, namely every curve at every prime except the two curves $x^3 - x^2 + x + 2$ and $x^3 + x^2 + x - 2$ at $p = 139$, both of discriminant -139, which the code excludes because they are not squarefree modulo 139;

(c) Theorem 2.1 against direct summation of (1.1) for every pair at five primes in $[500, 1000]$ (13,340 comparisons);

(d) relation (2.5) for every torsion pair at those primes (400 comparisons);

(e) an implementation check of the statistic at 256-bit primes: 2,000 uniform random residues, none of which may have $\theta_p \le 3/4$, and 2,000 planted lifts u/d with $|u| < 2^{140}$ and $d \le 10^4$, all of which must have $\theta_p \le 3/4$ with the lift recovered in reduced rational form.

Execution and disclosure. The run was carried out by the author on a Windows 11 laptop (CPython 3.12.10, python-flint 0.9.0, PARI/GP 2.19.0) and took 1,298 s, of which 1,151 s were point counts and 83 s the evaluations. Before release the package was smoke-tested on the torsion pairs only, at 128-bit primes and with separate seeds; no non-torsion pair of the grid was evaluated at a large prime before the run.

5 Results

Stage 0 passed with no mismatches in any of (a)–(d), no random residue flagged and no planted lift missed or misread. All 552 traces passed the consistency check; six of them were also recomputed afterwards with PARI/GP 2.15.4 on a different machine, with identical results. Table 1 summarizes the survey.

	Torsion pairs	Non-torsion pairs
Pairs	80	2,588
Classification	80 LIFT	2,588 NONE
θ_p at the three primes	≤ 0.519 (0 in 30 values)	≥ 0.9235 (median 0.968)
Recovered relation	80/80 as registered	none sought
Flagged for confirmation	—	0

Table 1: Survey at three 256-bit primes. The 30 values $\theta_p = 0$ are the ten pairs with $C_f(a) \equiv 0$ (Example 2.7).

The largest θ_p among torsion pairs was 0.508, 0.512, 0.519 and 0.512 for orders 2, 3, 5 and 6. By (3.3), $\theta_p \leq 0.519$ gives $S_p \leq p^{0.519}$ and $\theta_p \geq 0.9235$ gives $S_p > p^{0.9235}/3$. Since $\log_2 p \geq 255.5$ for all three primes, at each prime the minimal lift size $S_p(C_f(a))$ of every non-torsion pair exceeds that of every torsion pair with nonzero residue by a factor greater than $p^{0.4045}/3 > 2^{101}$; the ten torsion pairs with $C_f(a) \equiv 0$ have $S_p = 0$ and are excluded from this ratio. Assuming the computed residues are correct, the bound $\theta_p \geq 0.9235$ and Lemma 3.2 exclude $S_p \leq p^{3/4}$ for every evaluated non-torsion residue. This is a finite computational conclusion; a NONE classification close to the threshold would not by itself give the same exclusion, and nothing here bears directly on the asymptotic conjecture. The smallest of the 7,764 non-torsion values was 0.9235, comparable with the smallest value, 0.919, in a simulation of 7,764 uniform residues modulo the first prime; comparing minima alone does not establish agreement of distributions. Under the uniform model, (3.5) and a union bound put the probability of any accidental LIFT value among the 7,764 below $1.3 \cdot 10^{-13}$. A random sample of 40 pairs was re-evaluated independently from the saved traces, reproducing every θ_p. The verdict under the registered rules was that H4 is supported.

6 Discussion and limitations

Among the pairs tested, $C_f(a)$ had a small rational lift at the sampled primes exactly when R_a is torsion over $\mathbb{Q}$, and the lift was then the one Proposition 2.2 predicts. This is consistent with Conjecture 2.8, and with the heuristic that the Miller coefficient $\alpha_N(R_a)$ of a point of infinite order carries no small rational structure. It does not resolve the conjecture, which is an asymptotic statement about density. The two kinds of evidence are complementary: Theorem 2.5 proves the uniform statement for balanced relations, whereas the survey examined individual large primes, without a basis and without the balance condition. The torsion cases of order 5 and 6 are explicit instances of the proposition, not new theorems in any deeper sense.

Relation to Katz's universal extensions. The coefficient $\alpha_N(R_a)$ is an invariant already studied by Katz (2014). For an elliptic curve with a rational point P, Katz uses the differential $\omega_D = \frac{1}{2}\frac{y+b}{x-a}\frac{dx}{y}$, which is exactly our ϑ_P, and defines $\omega_p(P^\#) = N_p\,\omega_D - dg_p/g_p$ with $\mathrm{div}(g_p) = N_p(P) - N_p(O)$: the reduction of a lift $P^\#$ of P to the universal vectorial extension, multiplied by N_p. Katz shows that $\omega_p(P^\#) = (1-\mathcal{C})(\omega_D)$ in characteristic p, first for $p \nmid N_p$ (Katz, 2014, Lemma 8.1) and then in general, including $p \mid N_p$ (Katz, 2014, Remark 8.2 and Theorem 10.1, where $dg_p/g_p = (\mathcal{C} - a_p)\,\omega_D$ with $a_p = p + 1 - N_p$). Combined with Lemma 2.4, which is the same Cartier identity derived here from Theorem A of Jordan (2026), this gives $\omega_p(P^\#) = -\alpha_{N_p}(P)\,\omega_0 = -\frac{1}{2}\alpha_{N_p}(P)\,dx/y$; in Katz's basis dx/y, rather than our $\omega_0 = dx/(2y)$, the coefficient is therefore $c_p = -\alpha_{N_p}(P)/2$. Katz conjectures that, when P has infinite order, the normalized sequence $\omega_p(P^\#)/p$ is equidistributed in $\mathbb{R}/\mathbb{Z}$ (Katz, 2014, Conjecture 5.1), and he relates this to an older uniform torsion conjecture (Katz, 1972, Conjecture 7.5.11) under the assumption that $p \nmid N_p$ for all good p (Katz, 2014, Section 6); we do not claim that Theorem 2.5 is equivalent to that older conjecture in general. Katz's equidistribution conjecture would imply Conjecture 2.8. Indeed, by (2.7) a fixed relation (2.5) gives $\alpha_N(R_a) \equiv \kappa_1 + \kappa_2\varepsilon\tau_f$ with fixed rationals κ_1, κ_2. Choose a positive integer D such that $D\kappa_1/2$ and $D\kappa_2/2$ are integers k_1, k_2. Then D times the Katz coefficient $c_p = -\alpha_N(R_a)/2$ of (E_a, R_a) is congruent modulo p to $-(k_1 + k_2\varepsilon\tau_f)$, an integer of absolute value at most $|k_1| + 2|k_2|\sqrt{p}$ by Hasse's bound. Hence, at every prime where the relation holds, c_p/p lies within $(|k_1| + 2|k_2|\sqrt{p})/(Dp)$ of a point j/D modulo 1. For any $\delta > 0$ these points eventually lie in a fixed union of D intervals of total length δ, so equidistribution bounds the upper density of such primes by δ, for every δ. The modulo-p^2 elliptic Wieferich quotients of Silverman (1988) and Voloch (2000) are a different invariant; we have not analysed their relation to α_N, and nothing here identifies the two.

What made the survey possible. The survey depended on evaluating $C_f(a)$ at 256-bit primes. Direct summation of (1.1) would need about 10^{77} terms per value. The standard trace-free route through Bostan–Gaudry–Schost recurrences (Bostan, Gaudry & Schost, 2007) would need about 2^{128} field operations per value, with logarithmic factors suppressed. At an assumed 10^9 field operations per second that is about $1.1 \cdot 10^{22}$ years, and the usual square-root storage, about 2^{128} field elements, is about $1.1 \cdot 10^{40}$ bytes at 32 bytes per element. These are algorithm-specific estimates for that route, not lower bounds over all possible methods. We know of no standard package that evaluates $C_f(a)$ at an arbitrary pole. Semaev (1998) used a differentiated Miller recurrence to compute the logarithmic-derivative map on the p-torsion of elliptic curves in characteristic p, and Rück (1999) extended this to divisor class groups of curves of arbitrary genus. Theorem 2.1 builds on that framework but runs the chain at the full curve order $N = \#E_a(\mathbb{F}_p)$, for any trace. In the survey, standard point counting supplied the traces (1,151 s for 552 traces), and the differentiated evaluator then supplied all 8,004 sums in 83 s, about 10 ms each in pure Python, with each trace serving every pole of its curve. A search restricted to guessed quantities, testing (2.5) for fixed rationals, could have been run without Theorem 2.1, by direct summation at many primes near 10^6 and a relation search across them, as in the search that led to Theorem 6.3 of Jordan (2026). A basis-free size test at such primes would need many repeated evaluations per pair (Remark 3.4). The evaluator made a single-prime, basis-free test practical, one that would also have detected lifts involving quantities that no one thought to include in a basis.

The evidence has the following limits.

- *Independence.* Translations $x \mapsto x + c$ give identical sums, and $x \mapsto -x$ changes only the sign. The 2,588 non-torsion pairs give 1,834 distinct values up to sign at the three primes. The 184 curves show 54 distinct signatures ($|\tau_f|$) at the three primes; distinct signatures exclude equivalence up to isogeny and quadratic twist, but matching signatures at three primes do not establish it, and we have not classified the curves further.

- *Scope of the test.* Only rational representatives with small numerator–denominator product are detected, with the threshold $\theta_p \le 3/4$, which implies $S_p \le p^{3/4}$. A residue equal to the reduction of a small algebraic number in a fixed number field, or to a combination involving p-dependent quantities other than ε and τ_f with large coefficients, would not be seen.
- *Finite evidence.* Each pair was tested at three primes of one size, and the grid has small height. A NONE classification is strong evidence that no small lift exists at those primes, but proves nothing in general.
- *Modelling assumption.* The bound (3.5) is rigorous for uniform residues; that non-torsion sums behave like uniform residues is not proved. The 2,000-sample calibration checks the implementation and gross behaviour of the statistic; it cannot validate probabilities near 10^{-17}.
- *Novelty and review.* An AI-assisted literature review located the ingredients of Theorem 2.5 (Cartier, 1957; Katz, 1970; Bost, 2001) and the closely related work of Katz (2014) described above, but no explicit statement of Theorem 2.5 or of Conjecture 2.8; that review was not exhaustive, and the proof of Theorem 2.5 awaits specialist review.

The method is the main point. Where an evaluator reaches residues at primes of cryptographic size, size alone becomes a sharp, basis-free detector of hidden rational structure. The same test could be applied to the truncations $T_f(\mu)$ of Jordan (2026), to other stopping families, or to higher genus wherever comparable evaluators exist.

Status, AI assistance and data

This note is unrefereed. Generative AI was used extensively, under the author's direction, in designing the experiment, writing the code, checking results, reviewing the literature, assembling the proof of Theorem 2.5 from published theorems, and preparing the note. The code (`cm_round4`, building on `cm_explore`), its pre-registration and the complete run records, including traces, per-pair statistics and fingerprints, are available from the author; the evaluator and its paper are at https://github.com/bbrhuft/Cartier-Miller-evaluation-of-genus-one-coefficient-sums.

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